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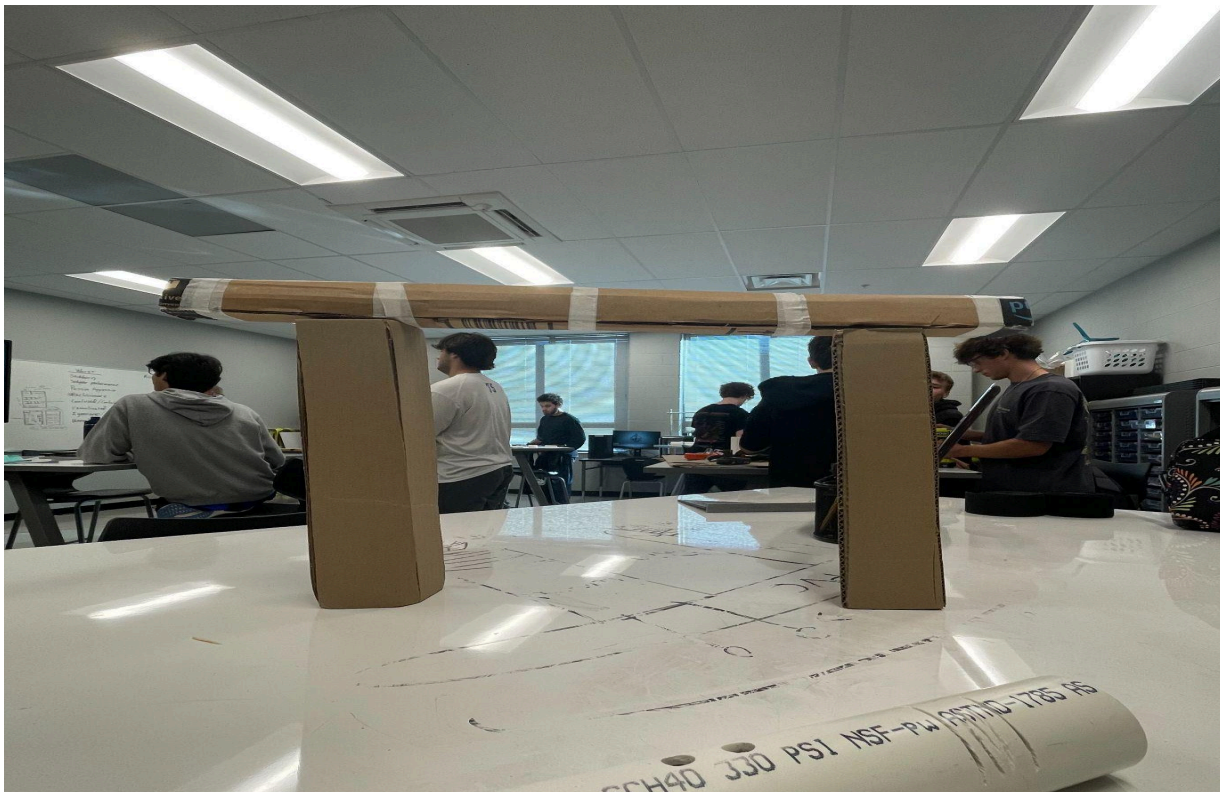
The primary problem that our group wanted to fix was the fact that Mr. Kiser experienced pain when he bent over. Our plan was to design an adjustable cart handle that could be efficient for users of all heights, allowing for a comfortable and easy experience.

We began by comprehending the problem and gathering relevant data. After we researched old attempts to fix this problem and analyzed them, we were finally allowed to brainstorm. While we brainstormed, each of us contributed ideas drawn from our prior knowledge and experience. We used methods from class such as common commons, which is the group's favorite ideas from a selection of each member's favorite personal ideas. This resulted in a wide choice of the most efficient potential solutions.

For a different perspective, Mr. Kiser had our class think of three other possible solutions to the issue, and also three ridiculous ideas as well. Some ideas that our group came up with were handles like those at the gym and stroller handle bars. In further depth, we chose these because they have an implemented component of ergonomics, which we needed in order to fulfil Mr. Kiser's wishes of a comfortable product.

The final design that we, as a group, agreed upon was a bar that extends from an extension of velcro. We came up with this by starting to design the setup of our product on the whiteboard table. We then brainstormed dimensions that we thought would be best suited for the materials and cart we were working with. With the look and dimensions of how we wanted the product to be, we also brainstormed potential materials. Considering the fact that our budget had to be under $30, we wanted either free or cheap, durable components. Colt was kind and offered the

robotics club's unused materials to the class. We then used wood and pvc, which we gathered would be durable and comfortable, from the robotics closet, and velcro from the classroom.

We took an extra step along the prototype making process, meaning that instead of jumping right into using the materials we planned on, we made the product out of cardboard first. We did this so that we could make sure we liked how the product looked and so we could make sure the dimensions that we previously picked were correct. To create the pvc and wood portions, we scored the cardboard with a box cutter which allowed for the cardboard to fold without us having to cut and glue it back together.

After we constructed the cardboard prototype, we decided that we liked how it looked and we then began on our actual product prototype. This process allowed for us to brainstorm with having knowledge first and not restrict ourselves with one idea.